

The Economics of Pharmaceuticals

A Plain-Language Guide

Why does the same drug cost \$48 in one country and \$292 in another? Why do payers, patients, and physicians see the price of a new therapy so differently? This chapter explains the economics behind pharmaceutical pricing, insurance, and the methods researchers use to measure whether a drug is worth what it costs.

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Based on Kevin A. Schulman, “Pharmacoeconomics: The Economics of Pharmaceuticals,” in *Pharmacoepidemiology*, 6th Edition, edited by Brian L. Strom, Stephen E. Kimmel, and Sean Hennessy

What This Chapter Covers

Modern medicine depends on pharmaceutical products, yet their price and economic impact on the healthcare system remain deeply controversial. This chapter explores the economics of the pharmaceutical market from discovery to the pharmacy counter.

We will look at how drugs get priced, how health insurance changes the way patients and manufacturers behave, a real-world pricing controversy over a hepatitis C cure, the research methods used to measure a therapy's value, a detailed case study of a landmark cancer drug, and how economic analysis translates into real purchasing decisions.

FOUNDATIONS, 2 OF 5

The Pharmacy Counter and the Market Behind It

A patient walks into a pharmacy with a prescription from their physician for a new medication. They have health insurance through their employer that covers part of the cost. The pharmacist dispenses the drug and collects the patient's share; the pharmacy bills the insurer for the rest.

This kind of transaction happens daily at every pharmacy in the United States and in similar forms elsewhere. Behind this simple, routine moment sits the complex economics of healthcare and the pharmaceutical market: how the drug came to exist, how its price was set, and how insurance changed what the patient actually pays.

How a Drug Gets Priced

Drug development moves through stages: discovery, optimization, clinical development, regulatory review, and launch. It often starts with publicly funded science, through the National Institutes of Health or the National Science Foundation, or private funders such as the Howard Hughes Medical Institute.

The gap between publicly funded basic science and the private investment needed to develop a molecule is called the “Valley of Death.” Discovering a candidate drug leads to a patent, giving the inventor the right to own the discovery. Clinical testing, regulated by the FDA, can take up to a decade and cost more than \$1 billion. At approval, the manufacturer sets a price reflecting its investment, the risk it took on, and considerations of market access.

FOUNDATIONS, 4 OF 5

The Price of Innovation, in Numbers

Prices for specialty pharmaceutical products can be extraordinary, and they vary enormously across countries for the very same medication.

\$475,000

Novartis's CAR-T cancer therapy, priced per patient.

Xarelto, 30-Day Supply

\$48 in South Africa, \$102 in Switzerland, \$292 in the United States.

Avastin, 400mg

\$956 in South Africa, \$1,752 in Switzerland, \$3,930 in the United States.

Oncology Drugs

Monthly cost rose from about \$100 in 1980 to \$10,000 by 2010.

From Patients' Pockets to Insurance

Historically, prescription drugs were relatively affordable and paid for directly by patients. As medications became more effective, prescription drug insurance began to develop and reshape the market.

In 1960, 96% of US prescription drug spending was out of pocket. By 1980 it was 71%, by 1990 it was 57%, by 2000 it was 28%, by 2010 it was 18%, and by 2015 it was down to 14%. Over the same period, the prescription drug market grew from \$2.7 billion in 1960 to \$462 billion in gross sales by 2016, yielding \$318 billion in net sales to manufacturers. Insurance works best when the insured event is infrequent, potentially catastrophic, and not influenced by the person being insured — conditions that make healthcare an unusually expensive insurance product to offer. In practice, most health insurance combines two different things: prepayment for routine, expected healthcare services, and protection against rarer, catastrophic costs — though some high-deductible plans try to separate the two.

HEALTH ECONOMICS, 1 OF 12

Health Economics: Moral Hazard and Behavior

Health economists have long studied how health insurance changes the way people consume healthcare. This section covers two frameworks for understanding that change: moral hazard, and behavioral economics.

Together, these ideas explain some of the most puzzling features of the pharmaceutical market: why prices can rise even as insurance is meant to protect patients, and why patients facing serious illness often behave very differently than economic theory would predict.

What Is Moral Hazard?

Moral hazard describes how individual behavior changes between paying for something yourself and having a third party pay. Kenneth Arrow won the Nobel Prize in economics for developing this framework, and Mark Pauly further developed the theory of demand.

We normally buy something when we believe it is worth at least what we pay for it. Third-party payment changes this calculus: when someone else is covering most of the cost, we may purchase things we would not have bought — or would have bought a cheaper version of — if we were paying the full price ourselves. This is not necessarily a bad thing: the same mechanism, sometimes called “good moral hazard,” can also let people obtain valuable care they could not otherwise have afforded.

The Value of a Dollar Changes With Insurance

Seven products, each with a different value to the patient, cost \$1.00 without insurance and just \$0.20 with a 20% co-insurance plan.

Metric	P1	P2	P3	P4	P5	P6	P7
Value	1.50	1.25	1.00	.75	.50	.25	.10
Cost, No Insurance	\$1.00	\$1.00	\$1.00	\$1.00	\$1.00	\$1.00	\$1.00
Cost, Insured	\$0.20	\$0.20	\$0.20	\$0.20	\$0.20	\$0.20	\$0.20

Without insurance, only products 1 through 3 — those worth at least their \$1.00 price — are worth buying. With insurance, the price drops to \$0.20, so products 1 through 6 now clear that bar. Insurance does not just lower the price; it changes which products look worth buying at all.

The Restaurant-Check Problem

A simple analogy illustrates the mechanism. Imagine going to dinner with friends and deciding in advance whether to pay individual checks or split the bill evenly.

With individual checks, you might order the cheaper salad since you are on a budget. But if the group decides in advance to split the bill, you may worry that everyone else will order the pricier steak — leaving you to pay your share of their expensive meals regardless. So you order the steak too, to at least get your share of the value. This is exactly how behavior changes between self-payment and third-party payment, and health insurance is one form of that third-party payment.

When Suppliers Respond to Moral Hazard

The discussion of moral hazard usually focuses on how it changes patient demand. But moral hazard also has a supply-side effect: sophisticated manufacturers can notice that insured patients only weigh their out-of-pocket cost, not the full price, and raise prices accordingly.

Suppliers have little power to influence prices in a competitive market. But healthcare often gives suppliers real pricing power — through a patent, for instance, or a product developed for a niche category too small to attract competition, such as an orphan drug. Under these conditions, a manufacturer can price a product based on the value patients perceive, not simply the cost of producing it.

Price, Value, and Market Power

Imagine a manufacturer choosing between two possible products, A and B, each costing \$50 million to develop. Product A targets a market of 20,000 patients; Product B, a smaller market of only 10,000.

Without Market Power

Assume both initially have the same \$10,000 market price. Product A's larger market brings in \$200M versus \$100M for Product B. The firm develops Product A.

With Market Power

Product B offers five times the value. Priced at \$50,000 to match it, B now brings in \$500M versus A's \$200M. The firm develops Product B instead.

Co-payment coupons and patient assistance programs, by lowering what patients feel they pay, can give suppliers even more room to raise prices this way.

Risk Aversion and Buying Insurance

Buying health insurance in the first place is consistent with people being risk averse: paying a set amount every year to avoid the potential financial shock of becoming severely ill.

But buying health insurance is not the same as buying healthcare. Whether the same risk-averse approach people take when purchasing insurance carries over into their treatment decisions once they are sick is a genuinely open question.

Gains, Losses, and Prospect Theory

Daniel Kahneman won the Nobel Prize in economics in 2002 for showing that people make different decisions under conditions of gain than under conditions of loss, a framework developed with Amos Tversky, who had passed away before the prize was awarded.

Under conditions of gain, we tend to be risk averse. Under conditions of loss, we tend to be risk seeking. A 70-year-old woman who refuses a flu shot over side-effect worries, while in a condition of full health, is acting under a condition of gain and being risk averse. This framework helps explain the very different, seemingly risk-seeking behavior of patients facing life-threatening disease.

A Lung Cancer Diagnosis

Consider an otherwise healthy patient who feels great, has a full social life, and has a lot to look forward to. A routine chest X-ray, ordered because of a smoking history, reveals a spot that turns out to be lung cancer that has already spread.

This otherwise healthy person now faces a life-threatening condition and a significant loss in life expectancy. Under this condition of loss, the patient will typically accept any treatment with a chance of restoring their health, regardless of its side effects — and often without asking about its cost. The way people make decisions changes entirely between how they felt buying their health insurance and how they feel confronting a life-threatening diagnosis.

Two Systems of Decision-Making

Kahneman and others have also focused on the role emotion plays in decision-making, describing two different ways people make decisions: System 1 and System 2.

System 1 is autonomous, fast, and represents our “gut” or emotional response to an uncertain situation. It easily incorporates societal attitudes but is subject to many systematic flaws. System 2 is more data-driven and analytical, but carries a higher cognitive burden. We normally rely on System 1 to minimize that burden, and in a heightened emotional state — such as facing a serious diagnosis — we generally rely on System 1 even for complex medical decisions.

Why Some Diseases Loom Larger Than Others

Public perception and communication around a diagnosis can carry significant emotional weight. Patients “fight” a cancer diagnosis, and society speaks of a “war” on cancer — even though heart disease is a far larger killer.

In 2013, heart disease was responsible for almost 300,000 deaths among women, while only about 40,000 women died of breast cancer that same year. Breast cancer is a terrible disease, but the emotional tone surrounding it is not simply a function of its mortality. In his book *The Emperor of All Maladies*, Siddhartha Mukherjee profiles Dr. Sidney Farber, who started the Jimmy Fund in 1948 to raise money for childhood leukemia research — deliberately framing cancer as an emotional cause. That strategy's legacy continues today, from the Cancer Moonshot to the Precision Medicine Initiative at NIH.

The Result: Price-Inelastic Demand

Both emotion and loss help explain treatment choices for patients with life-threatening conditions. The result is demand for therapies even where the risks seem to outweigh the benefits, or the benefits are modest given the price.

Under either decision-making framework, loss or emotion, patients are not actively required to consider cost — and cost might not even affect their decisions. This suggests demand for these therapies can be price-inelastic. It also suggests that mechanisms meant to address moral hazard, such as cost-sharing, would not be expected to influence demand from patients facing emotional or life-threatening conditions — and excessive cost-sharing in these cases might even be considered unethical.

HEPATITIS C, 1 OF 4

Clinical Example: The Hepatitis C Story

The development of a new generation of therapies for chronic hepatitis C infection was greeted as a significant scientific advance. But the payer community was in shock over the price.

This section follows sofosbuvir, one of these new hepatitis C cures, from its launch price through the international debate it triggered over what a cure is actually worth — and how that debate reshaped the pharmaceutical market that followed it.

An \$84,000 Price Tag

Sofosbuvir launched at an original price of \$84,000 per patient for a 12-week course of therapy — double that for 24 weeks. Given how common chronic hepatitis C infection is, treating the entire infected population could have doubled health insurance premiums in the United States for a year.

Untreated, hepatitis C can progress to cirrhosis, liver failure, or hepatocellular carcinoma, a fatal form of liver cancer, requiring hospitalization, a liver transplant if a donor organ was available, or death. Yet the clinical data behind the price were still early: there were no long-term outcome studies showing the therapy actually reduced these complications, only short-term trials measuring a surrogate marker of viral response. Gilead, the manufacturer, argued the price was justified by the drug's clinical benefit — though a Senate investigation found that argument was absent from the company's own internal discussions in setting the price.

HEPATITIS C, 3 OF 4

Putting a Price on Benefit: NICE and the QALY

In the United Kingdom, a new therapy requires a formal economic evaluation of its impact on healthcare costs and outcomes before it can be added to the national formulary — assessing the incremental cost of the new therapy against the incremental clinical benefit it provides, expressed as a cost per quality-adjusted life-year, or QALY.

In the UK, a cost of £20,000 per QALY is considered a benchmark of good value. A detailed evaluation by the National Institute for Health and Care Excellence (NICE), considering a price of £34,504 per course, concluded that sofosbuvir had a high likelihood of meeting that £20,000 threshold for some indications — and recommended that the National Health Service find the resources to fund it.

How the Story Played Out

In its first year on the US market, sofosbuvir generated \$10 billion in revenue for Gilead, making it one of the top-grossing pharmaceutical products in the world. But the entry of competing hepatitis C therapies soon drove prices down, with treatment courses ranging from \$26,400 to \$62,500 by 2017.

Gilead's hepatitis C product sales fell from \$19.1 billion in 2015 to \$14.8 billion in 2016 as competition intensified. The pricing of sofosbuvir represents an interesting dilemma in modern health economics: the company could not have set such a high price in the absence of insurance coverage, yet that high price also helped patients receive better coverage, since it turned hepatitis C into a catastrophic medical expense rather than a routine one — and physicians and patients came to see it as a potentially life-threatening condition demanding treatment.

METHODS, 1 OF 11

How We Study Value: The Methods of Pharmacoeconomics

Given stories like sofosbuvir's, how should patients, physicians, and payers decide what a therapy is actually worth? Pharmacoeconomics is the field built to answer that question.

This section reviews how economic studies of pharmaceuticals are designed and timed, the core concepts economists use to define a “cost,” the different types of economic analysis, the different kinds of costs that get counted, and the perspective from which an analysis is conducted.

METHODS, 2 OF 11

Pharmacoeconomic Studies Serve Different Audiences

Pharmacoeconomic studies are designed to meet the different information needs of healthcare purchasers and regulatory authorities. Early economic data from Phase III trials support initial pricing and education; postmarketing studies compare new therapies with existing ones.

No single study can provide complete economic information to every interested audience. Studies are often designed for a specific perspective, such as a health maintenance organization, or to assess a specific cost category, such as productivity. Around the globe, technology assessment agencies evaluate this evidence: NICE in the UK, the Institute for Quality and Efficiency in Health Care (IQWiG) in Germany, and the Institute for Clinical and Economic Review (ICER), a private organization, in the United States.

METHODS, 3 OF 11

Building Economics Into Drug Development

The drug development process allows for the timely collection of data to evaluate a therapy's costs and effects early in its life, alongside the opportunity for further data collection once the product is approved and marketed.

Phase I and II studies develop pilot economic data, such as estimates of cost and quality of life, and pilot-test data collection tools. Phase III studies can include economic assessments as a primary or secondary endpoint, such as changes in hospitalization rates. After approval, postmarketing “pragmatic” or “practical” trials compare a therapy against usual care in more realistic, less controlled settings. Integrating economic data into a clinical trial requires designing the economic protocol early enough that it does not distort the clinical trial itself — for instance, by requiring different resource use across treatment arms in a way that biases the results.

Why Trial Economics May Not Travel

Economic analysis faces a generalizability challenge that runs in two directions, depending on the size and reach of a product's clinical trial.

Trials Too Small

Specialty pharmaceutical products are often tested in very small trials. Eteplirsen, an orphan drug for Duchenne muscular dystrophy, was tested in just a dozen patients, measuring dystrophin production and a six-minute walk test, with no long-term outcome data available.

Trials Too Global

Primary care products, such as treatments for hypertension or diabetes, are tested as large, global undertakings — raising the opposite question of whether pooled economic results actually generalize to a particular country or an individual payer's population.

What Economists Mean by “Cost”

Economists emphasize that costs are more than just transactions of currency. A cost represents the consumption of a resource that could otherwise be used for another purpose — what economists call the resource's “opportunity cost.”

For example, the time it takes to read this chapter is a cost to the reader, because that time cannot be used again; the opportunity to use it another way has been given up. It is also important to remember that not every transaction of money is a cost to society as a whole. Social security payments, disability payments, and similar transfers do not represent the consumption of resources — they simply transfer the right to consume resources from one person to another.

Three Dimensions of Economic Analysis

Economic analysis of medical care can be thought of as a cube with three dimensions: the type of analysis performed, the perspective taken, and the type of costs and benefits counted.

Type of Analysis

Cost-identification, cost-effectiveness, or cost-benefit.

Perspective

Society, patient, payer, or provider.

Costs and Benefits

Direct medical, direct nonmedical, productivity, or intangible.

The next several slides walk through each of these three dimensions in turn.

Cost-Benefit Analysis

Cost-benefit analysis compares the cost of a medical intervention to its benefit, with both measured in the same monetary units. This determines either the ratio of dollars spent to dollars saved, or the net saving or cost — an investment should be made when its benefits exceed its costs.

Analysts generally prefer comparing the incremental cost and benefit of a new therapy against conventional care, rather than comparing their totals, because it isolates the difference that actually matters. But an incremental analysis can overstate a new therapy's value if it is compared against an already expensive, less-effective option; a zero-based budgeting approach, which starts fresh from a base case, makes the true costs and benefits more transparent. The central difficulty is that it requires researchers to express clinical outcomes in monetary terms — translating years of life saved, or morbidity avoided, into dollars raises difficult methodologic and ethical questions about valuing human life differently across patients.

Cost-Effectiveness Analysis

Cost-effectiveness analysis avoids the dilemma of cost-benefit analysis. Cost is still calculated in monetary terms, but effectiveness is measured independently in clinical units, such as lives saved or a change in blood cholesterol. When different clinical outcomes are combined using patient utility weights, with the result expressed as QALYs, this specific approach is called cost-utility analysis.

Outcomes Improve, Costs Increase

Cost-effective: judged by benefit added per additional dollar spent.

Outcomes Improve, Costs Decrease

Cost-saving, or dominant: adopt without hesitation.

Outcomes Worsen, Costs Increase

Dominated: should never be adopted.

Outcomes Worsen, Costs Decrease

Cost-effective: judged by money saved per unit of benefit lost.

Net benefit combines incremental cost and effectiveness using a willingness-to-pay threshold, such as \$100,000 per QALY; a positive net benefit means the program should be adopted from a cost-effectiveness perspective.

Cost Identification: the Simplest Approach

An even less complex method than cost-benefit or cost-effectiveness analysis is to simply enumerate the costs involved in medical care and ignore the outcomes that result — an approach called cost identification analysis.

Cost identification studies compare treatments based on their costs alone, so they are only appropriate when the outcomes or benefits of the treatments being compared are equivalent. Because these studies, like the others, often rely on models built from imperfect data, researchers also run sensitivity analyses: recalculating the results using alternate values for uncertain variables, to test how much the study's conclusions depend on any one assumption.

Four Types of Cost

Direct Medical

Payments for drugs, physicians' fees, allied health salaries, and diagnostic tests.

Direct Nonmedical

Transportation, special clothing or housing needs, and hotel stays for distant care — usually paid out of pocket.

Productivity

Lost work time or premature death removing someone from the workforce — sometimes larger than direct costs.

Intangible

Pain, suffering, and grief. Difficult to measure, and often omitted from economic studies.

Whose Perspective Are We Taking?

Costs and benefits can be measured from the perspective of society, the patient, the payer, or the provider — and the choice of perspective determines how costs and benefits are measured.

A hospital's cost of providing a service may be lower than what it charges; from the patient's perspective, the full charge may be the more accurate reflection of their cost. It has been recommended that, as a base case, economic analyses adopt a societal perspective, since costs may not be borne by the same parties who benefit from care. Because perspective changes the answer, economic analyses of medical care regularly raise real ethical questions about equity, the distribution of resources, and who is responsible for a society's health.

IMATINIB CASE STUDY, 1 OF 8

Case Study: Imatinib and Chronic Myeloid Leukemia

The previous sections covered the principles of clinical economics and the methodologic issues in assessing pharmaceutical value. This section applies those methods to a real case: imatinib, marketed as Gleevec, a landmark therapy for chronic myeloid leukemia.

We will look at the disease itself, the discovery of a drug that targeted its underlying biology, the pivotal clinical trial that tested it, the economic model built to evaluate it, and why that economic analysis turned out to be surprisingly controversial.

What Is CML?

Chronic myeloid leukemia (CML) is a malignant disorder of blood-forming stem cells, accounting for 15 to 20% of adult leukemia cases.

The disease typically presents in a relatively symptom-free chronic phase, but can progress through accelerated and blast phases. In the chronic phase, median life expectancy is about eight years; in advanced phases, without treatment, that drops to just months.

IMATINIB CASE STUDY, 3 OF 8

From a Broken Chromosome to a Targeted Drug

CML results from a translocation between chromosomes 9 and 22, creating what is known as the “Philadelphia chromosome,” found in most patients with the disease.

This translocation forms a new gene, BCR-ABL, which produces a unique protein, a tyrosine kinase, that activates signals for uncontrolled growth of white blood cells. A collaboration between industry and academia searched for a kinase inhibitor that could target this protein directly, eventually discovering the molecule that became imatinib, marketed by Novartis as Gleevec. Economic evaluation was an early consideration in its development: from the sponsor's perspective, CML represented a limited US market of only about 5,000 patients — though if the drug worked as well as hoped, its clinical benefit could support a high price.

IMATINIB CASE STUDY, 4 OF 8

The IRIS Trial

The International Randomized Interferon versus STI571 Study, or IRIS, was an open-label trial comparing imatinib against interferon-alpha plus low-dose cytarabine (IFN+LDAC), the prior standard of care, in 1,106 newly diagnosed chronic-phase CML patients across 16 countries.

Discontinued First-Line Therapy

12.3% imatinib versus 31.6% IFN+LDAC.

Complete Cytogenetic Response at 18 Months

76.2% imatinib versus 14.5% IFN+LDAC.

Progressed to Accelerated Phase or Blast Crisis

3.3% imatinib versus 8.5% IFN+LDAC.

Crossed Over to the Other Treatment

2.0% imatinib versus 57.5% IFN+LDAC.

That heavy, one-directional cross-over — more than half the comparison group eventually switching to imatinib — substantially weakened a standard intention-to-treat analysis's ability to detect imatinib's true effect.

IMATINIB CASE STUDY, 5 OF 8

Modeling Survival Around a Crossover Problem

Given the extensive cross-over in IRIS, the economic analysis required building a model rather than simply reading survival off the trial data. Researchers modeled two treatment pathways: patients on imatinib who progressed switched to IFN+LDAC and then to hydroxyurea; patients on IFN+LDAC who progressed switched directly to hydroxyurea. The model also took a healthcare-system perspective, counting only direct medical costs, and converted quality-of-life responses collected with the EQ-5D questionnaire into the utility weights needed to calculate QALYs.

Because the two-year survival rate for the IFN+LDAC arm could not be reliably estimated from IRIS itself, given the cross-over, researchers instead used a historical control from an earlier Italian trial. They also used complete cytogenetic response (CCyR) — the disappearance of the Philadelphia chromosome from a patient's blood — as a biomarker of treatment effectiveness, modeling long-term survival from the increased risk of death among CCyR and non-CCyR patients compared with an age- and gender-matched general population, rather than the more typical approach of extrapolating a survival curve forward year by year.

What the Economic Model Found

In the base-case analysis, patients receiving imatinib were estimated to survive an average of 15.30 years, compared with 9.07 years for patients receiving IFN+LDAC — an incremental gain of 6.23 years, a very large effect relative to most pharmaceutical comparisons.

Survival Gain

6.23 additional years, or 5.85 additional quality-adjusted life-years (QALYs) after adjusting for quality of life.

Undiscounted Lifetime Cost

\$424,600 for imatinib versus \$182,800 for IFN+LDAC, a difference of \$241,800.

That cost difference reflects both the price of the medication itself and the cost of the additional years of treatment and life imatinib provided.

The Price of an Extra Year

Applying a standard 3% annual discount rate, which reflects that benefits received today are worth more than the same benefits received in the future, the incremental survival gain fell to 3.93 years and 3.89 QALYs, and the incremental cost fell to \$168,100.

Combining these discounted estimates produced incremental cost-effectiveness ratios, or ICERs, of \$43,100 per life-year saved and \$43,300 per QALY saved. Simulations of 1,000 patients showed a relatively low level of variability in these results and a high correlation between the incremental differences in cost and survival — with the upper limit of the 95% confidence interval for both ratios still under \$51,100. A tornado diagram, testing the effect of varying each assumption individually, confirmed the result was reasonably stable across a range of alternative assumptions about costs, survival, and treatment patterns.

Why This Analysis Was Controversial

This analysis was very controversial for two reasons. First, its entire survival benefit was a modeling exercise — that specific benefit had not actually been observed in the original IRIS trial itself, though the underlying relationship between cytogenetic response and survival was later confirmed in a five-year follow-up of the study.

Second, the economic analysis was based on imatinib's initial launch price. The manufacturer later raised the price well above that launch price over time, from \$26,400 to over \$120,000. That price also varied enormously by market: by 2013, costs in Canada were less than 20% of US costs, and a year's supply cost just \$2,500 in India. Patent expiry is expected to substantially reduce the drug's cost in the US, with the potential to bring the price down close to the actual cost of its active pharmaceutical ingredients.

THE FUTURE, 1 OF 7

The Future: From Analysis to Decisions

Health economics helps us understand both the supply and demand for pharmaceutical products, and pharmacoeconomics provides insight into the value of a product to patients, payers, and the marketplace. The next challenge is relating the results of that analysis to actual purchase decisions.

Should a patient, hospital, or public or private payer approve payment for a therapy — for example, by adding it to an approved formulary? This section looks at how that decision actually gets made, and the budget realities that constrain it.

The Easy Decisions, and the Hard Ones

If a product is thought to add clinical benefit and save money, it is an easy decision to add it — these products are described as cost-saving or dominant. Vaccines and generic drugs often fall into this category.

If a product worsens clinical outcomes and raises cost, it is also an easy decision — simply do not add it; these products are dominated, though this situation is much less common. The real, difficult challenge in formulary decisions is the vast majority of new therapies that add clinical benefit at additional cost, landing somewhere in between.

A Budget Question and a Value Question

Outside of healthcare, we make this kind of decision frequently. Imagine a \$100 grocery budget for your shopping list. A new, pricier item on the shelf raises two separate questions.

The first is a budget question: do you have the money to make this purchase at all? If the item is far outside your budget, the question becomes moot regardless of how good it is. The second, once you do have room to spend a bit more, is a cost-effectiveness question: given the increase in cost, which of two options provides more value? Healthcare works the same way — a therapy first has to clear a budget constraint before its cost-effectiveness even becomes the relevant question.

When Clinical Potential Meets Budget Reality

Budget constraints are built into real healthcare spending. National health insurance programs often work from a fixed annual allocation, and private insurers must estimate premiums for the coming year, sometimes as much as 18 months before the spending actually happens.

Sofosbuvir illustrates the problem starkly: at its \$84,000 list price and a potential US market of 3.2 million people with hepatitis C, treating everyone would have required a staggering \$269 billion budget. So while the clinical potential of the therapy was tremendous, the budget constraint itself limited how quickly it could be adopted. Some budget constraints are especially rigid: Medicaid, jointly funded by federal and state governments, is not legally allowed to run a budget deficit, so a state funding a costly new therapy must either raise revenue or cut spending elsewhere, such as in education.

The “Benefit Pool” View

Another way to think about the budget question looks at the impact on the health insurance premium of everyone in the pool, not just the funding agency's overall budget. This is called the “benefit pool” perspective.

For example, adding PCSK-9 inhibitors, used to lower cholesterol and priced at more than \$14,000 annually per patient, was calculated to add \$140 to the insurance premium for every person in the pool under modest adoption assumptions. More broadly, health insurance premiums have been shown to rise about \$250 for every 0.25% of a population that receives a \$100,000 drug for any indication — meaning even a fairly rare condition can meaningfully affect what everyone else pays for coverage.

Good Value Does Not Always Mean Affordable

Does good value for money automatically mean a therapy should be added to a formulary? The honest answer is: it depends. Cost-effectiveness analysis assesses the relative value of different therapies well, but that ratio is not tied to a product's budget impact.

A product that is not good value for money for a rare condition may still have a fairly modest budget impact, while a drug that is good value for money for a common condition could still have a very significant one. A product's value can also change by indication — patients with existing heart disease, for instance, are at higher risk than those without it, so treating them can provide more value per dollar than using the same therapy for primary prevention. Pricing by indication, to reflect this, has proven difficult to implement in practice. Rather than a single absolute threshold, the UK instead uses an implicit relative standard: products with a lower cost-effectiveness ratio are simply more likely to be recommended by NICE.

Saving Money at a Small Cost to Outcomes

There is one more quadrant worth considering: products that save money but come at some expense to clinical outcomes. This trade-off is actually fairly common in practice, as long as the money saved is large relative to the health benefit given up.

Amoxicillin, for example, is recommended as first-line therapy for otitis media (ear infections) despite a high level of antibiotic resistance, because of its low cost and the low likelihood of serious complications if this first, inexpensive treatment fails. This is also, in its own way, a values-based decision about the price of a small amount of clinical risk.

Key Takeaways

A drug's price reflects far more than the cost of producing it. It reflects the enormous risk and cost of development, the market's size, the value patients place on the therapy, and — because of insurance — a supply-side moral hazard that lets sophisticated sellers price to that perceived value rather than to cost.

Insurance changes behavior on both sides of the market. Patients facing loss, emotion, or a life-threatening diagnosis often make decisions with little regard to cost, producing demand that is largely price-inelastic — exactly the condition under which prices can rise furthest.

Measuring whether a therapy is “worth it” requires choosing a type of analysis, a perspective, and a set of costs to count — and even a therapy proven to be good value for money can still be unaffordable, because cost-effectiveness and budget impact are two different questions entirely.

Thank You

The price of pharmaceutical products is an important challenge for everyone in the healthcare value chain: patients, physicians, payers, and government. We all want innovation in healthcare — but it is not yet clear whether we can afford that innovation at any price.

As reimbursement decisions become more centralized, they increasingly call for collaboration between epidemiologists and economists — a merger that can put better information in front of decision makers and use limited resources more effectively for the health of the public.

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